

JORDAN HOURI

943 Paca St
Indianapolis, IN 46202

Email: jordanhouri@gmail.com

Phone: (858) 354-1873

EDUCATION

- 08/2024 – Present **MD – *Indiana University***, Indianapolis, Indiana
- 08/2019 – 05/2021 **MS – *Duke University***, Durham, North Carolina
Medical Physics
Thesis: “A General-Purpose Simulator for Evaluating Astronaut Radiation Exposure”
Advisors: Anuj Kapadia, PhD [anuj.kapadia@duke.edu]
Ehsan Samei, PhD [esi.samei@duke.edu]
- 10/2015 – 06/2018 **BA (Honours) – *University of Oxford***, Oxford, UK
Physics

PROFESSIONAL EXPERIENCE

- 05/2026 – Present **Research Assistant – *Department of Electrical & Computer Engineering, Purdue University***, West Lafayette, Indiana
- Contributed to the development of physics-informed constrained optimization methods for mechanical strain tensor reconstruction from hyperspectral neutron computed tomography data as part of a collaboration with Oak Ridge National Laboratory.
- Supervisors: Charles Bouman, PhD [bouman@purdue.edu]
Gregory Buzzard, PhD [buzzard@purdue.edu]
- 05/2022 – 05/2024 **Lead Scientist, Space Exploration – *StemRad, Inc.***, Tampa, Florida / Tel Aviv, Israel
- Matroshka AstroRad Radiation Experiment (MARE)
 - Led StemRad’s scientific evaluation of the effectiveness of the AstroRad vest in deep space during NASA’s Artemis 1 lunar mission, during which two identical radiation dosimetry phantoms, *Helga* and *Zohar*, orbited the Moon onboard the Orion capsule, with *Zohar* wearing an AstroRad vest.
 - Comfort and Human Factors AstroRad Radiation Garment Evaluation (CHARGE)
 - Improved the design of the AstroRad vest based on feedback regarding ergonomics, range of motion, comfort and general user experience from astronauts who evaluated a prototype vest onboard the International Space Station.
 - Managed a collaboration between StemRad and Redwire Space to develop the capability to recycle plastic waste into 3D printer filament for additive manufacturing of AstroRad shielding cores in zero gravity.
- CEO: Oren Milstein, PhD [oren@stemrad.com]
- 08/2021 – 04/2022 **Computational Scientist, Radiation Biology & Monte Carlo Modeling – *Advanced Computing for Health Sciences, Oak Ridge National Laboratory***, Oak Ridge, Tennessee

Developed a GEANT4-based computational framework to embed microscopic multicellular models inside voxelized digital human body phantoms, and helped incorporate it into an automated pipeline to model fractionated irradiation and tumor cell proliferation over time.

Supervisors: Anuj Kapadia, PhD [kapadiaaj@ornl.gov]

Greeshma Agasthya, PhD [agasthyaaga@ornl.gov]

08/2019 – 05/2021 **Graduate Student Researcher – *Carl E. Ravin Advanced Imaging Laboratories (RAI Labs), Duke University Department of Radiology***, Durham, North Carolina

Conducted master's thesis research on the development of computational models for predicting human radiation exposure during space exploration missions at any location in the solar system.

Supervisor: Anuj Kapadia, PhD [anuj.kapadia@duke.edu]

07/2018 – 08/2019 **Postbaccalaureate Research Fellow – *Department of Radiation Medicine, University of California, San Diego***, San Diego, California

Developed convolutional neural network models to automatically segment brain metastases in T1 MRIs and track them across multiple scans for stereotactic radiosurgery (SRS) patients.

Supervisors: Jona Hattangadi-Gluth, MD [jhattangadi@health.ucsd.edu]

Albert Hsiao, MD, PhD [a3hsiao@ucsd.edu]

07/2017 – 10/2017 **Undergraduate Research Volunteer – *Department of Radiation Medicine, University of California, San Diego***, San Diego, California

Developed a computational pipeline to analyze local radiation-dose-dependent changes in diffusion-weighted MRI biomarkers along the central axes of white matter tracts in the brain.

Supervisor: Vitali Moiseenko, PhD [vmoiseenko@health.ucsd.edu]

SERVICE

09/2022 – 05/2024 **Volunteer – *Moffitt Cancer Center Radiation Therapy Unit***, Tampa, Florida

Assisted patients with finding their way from the lobby to the treatment room, provided warm blankets, and helped comfort and reassure them as well as their families.

Volunteer Coordinator: Jessica Collard [jessica.collard@moffitt.org]

11/2021 – 04/2022 **Volunteer – *Parkwest Medical Center Emergency Department***, Knoxville, Tennessee

Aided with check-in, restocking supplies, patient transportation, and assisting patients by providing water and blankets.

Volunteer Coordinator: Charlene Howard [chowar10@covhlth.com]

04/2020 – 07/2021 **Volunteer – *Meals on Wheels Durham***, Durham, North Carolina

Delivered meals to elderly people who can no longer shop or cook for themselves during the COVID-19 pandemic.

Director: Annette Read [annette@mowdurham.org]

06/2020 – 01/2021 **Volunteer Intern – *Durham Partnership for Seniors – COVID Response Digital Divide Task Group***, Durham, North Carolina

Worked on a community assessment on technology and broadband internet or wireless service availability to improve access to food and healthcare and decrease social isolation among older adults and persons with disabilities.

Faculty Mentor: Janet Prvu Bettger, ScD, MS [janet.bettger@duke.edu]

04/2020 – 08/2020 **Organizer, Volunteer – *Duke Medical Physics Graduate Program Face Shield Initiative***, Durham, North Carolina

Organized a group of Medical Physics graduate students to construct DIY reusable plastic face shields, which were donated to Durham hospitals and clinics at the start of the COVID-19 pandemic.

11/2008 – 08/2019 **Volunteer – *Loving Care Adult Day Health Center***, San Diego, California

Aided in physical and emotional healthcare of older adults and persons with disabilities.

Administrator: Yelena Frolova [(619) 718-9777]

06/2012 – 09/2015 **Founder, Volunteer – *Connective Living***, San Diego, California

Launched a local initiative to team up teenagers with elderly immigrants to help bring them online and become familiar with social media for improving connections with friends, family, and caregivers.

SKILLS

Programming Languages: Python, C++, MATLAB, CUDA, Fortran.

Software / APIs: GEANT4, MBIRJAX, Tensorflow.

Miscellaneous: High-performance computing, particle physics, radiation dosimetry, CT reconstruction, Monte Carlo modeling, neural networks, CAD modeling & 3D printing.

PROFESSIONAL ACTIVITIES

08/2024 – 05/2025 Research Chair – *IUSM Aerospace Medicine Student Interest Group*

CERTIFICATIONS

08/2023 ***NASA Human Research Program, Space Radiation Element***
SHINE Space Radiation Training Program

HONORS / AWARDS

- | | |
|--|----------|
| • Jack Bauer Honor Fund Conference Travel Award – <i>Indiana MSTP</i> | Jun 2024 |
| • Haselby Scholarship – <i>Indiana MSTP</i> | May 2024 |
| • Dean's Research Award – <i>Duke University Graduate School</i> | Oct 2020 |
| • Zachary D. Shrock Memorial Scholarship – <i>Duke Medical Physics Graduate Program</i> | May 2020 |
| • Academic and Research Excellence Award – <i>Duke Medical Physics Graduate Program</i> | May 2020 |
| • Old Members' Trust Conference & Academic Travel Award – <i>University College Oxford</i> | Apr 2018 |
| • 2015 Academic Hall of Fame – <i>Westview High School</i> (5 inductees of 560 graduates) | Jun 2015 |
| • U.S. Presidential Scholars Candidate | Jan 2015 |
| • National Merit Scholarship | Sep 2014 |

TEST SCORES

- MCAT – 520 (*C/P 132, CARS 129, B/B 128, P/S 131*) Mar 2022
- GRE – 334 (*Verbal 164, Quantitative 170*) Dec 2017
- ACT – 36 (*All subsections*) Apr 2014

PUBLICATIONS

1. **Houri J**, Berger T, Gaza R, Hussein H, Schwarz S, Waterman G, Matthiä D, Marsalek K, Przybyla B, Aeckerlein J, Radenhausen M, Kasemann M, Coderre K, Segars W P, Samei E, Kapadia A, Lee K, Laramore D, George S, Semones E, Milstein O. [First Evaluation of Wearable Radiation Protection for Human Deep Space Exploration, as Flown on Artemis I](#). *Science Advances*, (2026). – **Featured on journal cover.**
 ↳ Highlighted in: [Research highlights – “Astronaut vest cuts radiation exposure in half”](#). *Nature*, (2026).
2. Truong E, **Houri J**, Hermann G, Moiseenko V, Hattangadi-Gluth J, Rudie J, Simon A. [Effect of Anatomical Contextual Information on the Performance of a Convolutional Neural Network Tasked with Brain Metastasis Detection](#). *Advances in Radiation Oncology*, (2026).
3. George S, Gaza R, Matthiä D, Laramore D, Lehti J, Campbell-Ricketts T, Kroupa M, Stoffle N, Marsalek K, Przybyla B, Abdelmelek M, Aeckerlein J, Bahadori A, Barzilla J, Dieckmann M, Ecord M, Egeland R, Eronen T, Fry D, Jones B, Hellweg C, **Houri J**, Hirsh R, Hirvonen M, Hovland S, Hussein H, Johnson A, Kasemann M, Lee K, Leitgab M, McLeod C, Milstein O, Pinsky L, Quinn P, Riihonen E, Rohde M, Rozhdestvenskiy S, Saari J, Schram A, Straube U, Turecek D, Virtanen P, Waterman G, Wheeler S, Whitman K, Wirtz M, Vandewalle M, Zeitlin C, Semones E, Berger T. [Space Radiation Measurements During the Artemis I Lunar Mission](#). *Nature*, (2024).
4. Salans M, **Houri J**, Karunamuni R, Hopper A, Delfanti R, Seibert T, Bahrami N, Sharifzadeh Y, McDonald C, Dale A, Moiseenko V, Farid N, Hattangadi-Gluth J. [The Relationship Between Radiation Dose and Bevacizumab-Related Imaging Abnormality in Patients with Brain Tumors: A Voxel-Wise Normal Tissue Complication Probability \(NTCP\) Analysis](#). *PLOS ONE*, (2023).
5. **Houri J**, Karunamuni R, Connor M, Pettersson N, McDonald C, Farid N, White N, Dale A, Hattangadi-Gluth J, Moiseenko V. [Analyses of Regional Radiosensitivity of White Matter Structures along Tract Axes using Novel White Matter Segmentation and Diffusion Imaging Biomarkers](#). *Physics and Imaging in Radiation Oncology (phiRO)*, (2018).

CONFERENCE ABSTRACTS

1. **Houri J**. Fast Superposition Algorithm Using Spectral Kernel Decomposition for Photon Beam Dose Calculation in Heterogeneous Media. *2026 Joint AAPM | COMP Annual Meeting*, Vancouver, Canada (2026).
2. **Houri J**, Milstein O, Berger T, Gaza R, Hussein H, Schwarz S, Waterman G, Marsalek K, Matthiä D, Przybyla B, Aeckerlein J, Coderre K, Segars P, Samei E, Kapadia A. First Data-Based Evaluation of the Radiation Protection Capabilities of the AstroRad Vest as Flown Onboard Artemis I. *75th International Astronautical Congress (IAC)*, Milan, Italy (2024). – **Oral Presentation.**
3. **Houri J**, Milstein O, Schwarz S, Lazar S, Waterman G, Chowdhury R, Fleisher T, Kloske J, Cartolano M, Schulman H, Bussler V, LeMand R, Thompson M, Longo A, Shestopole P, Flowers P, Piness J, Shaked Y, Mulligan M, Savin K. Recycling of On-Orbit Plastic Waste into Personal Radiation Protection using Additive Manufacturing. *13th Annual International Space Station Research and Development Conference (ISSRDC)*, Boston, Massachusetts (2024). – **Oral Presentation** (Presented by Ken Savin due to scheduling conflict).
4. Milstein O, **Houri J**, Coderre K, Schwarz S, Chowdhury R, Lazar S, Morgan E, Hussein H. The Comfort and Human factors AstroRad Radiation Garment Evaluation (CHARGE) Onboard the ISS. *12th Annual International Space Station Research and Development Conference (ISSRDC)*, Seattle, Washington (2023). – **Oral Presentation.**

5. Berger T, Przybyla B, Aeckerlein J, Marsalek K, Matthiae D, Wirtz M, Rohde M, Kasemann M, Hellweg C, Gaza R, Lee K, Milstein O, **Houri J**, Hussein H. The MARE Experiment On-Board the NASA Artemis I Mission. *2023 NASA Human Research Program Investigators Workshop (HRP IWS)*, Galveston, Texas (2023).
6. Hermann G, **Houri J**, Connor M, Karunamuni R, Hsiao A, Noorbakhsh A, Simon A, Seibert T, Farid N, Rudie J, Hattangadi-Gluth J. [3D Convolutional Neural Network for Automated Segmentation of Intracranial Metastases and Organs at Risk for Brain SRS: Implications for Treatment Planning and Longitudinal Tracking](#). *64th American Society for Radiation Oncology (ASTRO) Annual Meeting*, San Antonio, Texas (2022).
7. Inman P, **Houri J**, Gounley J, Agasthya G, Kapadia A. A Computational Multiscale Framework for Precision Dosimetry in Radiation Therapy. *64th American Association of Physicists in Medicine (AAPM) Annual Meeting*, Washington, DC (2022).
8. Lee H, Agasthya G, Hanson H, Logan J, **Houri J**, Kapadia A, Dewji S. Machine Learning Methods and Multivariate Epidemiology in Radiation Risk Assessment Models. *67th Health Physics Society (HPS) Annual Meeting*, Spokane, Washington (2022).
9. **Houri J**, Segars P, Samei E, Kapadia A. A General-Purpose Simulator for Evaluating Astronaut Radiation Exposure. *67th Radiation Research Society (RRS) Annual Meeting*, Online (2021).
10. **Houri J**, Kapadia A. A Scatter-Rejection Method for Proton Computed Tomography. *63rd American Association of Physicists in Medicine (AAPM) Annual Meeting*, Online (2021).
11. **Houri J**, Segars P, Kapadia A. Monte Carlo Calculation of Radiation Exposure to Astronauts Using 4D Extended Cardiac-Torso (XCAT) Phantoms. *2020 Joint AAPM/COMP Virtual Meeting*, Online (2020).
12. **Houri J**, Noorbakhsh A, Bahrami N, Karunamuni R, Hsiao A, Hattangadi-Gluth J. [Met-SRSNet: An Ensemble Approach to Automated Detection and Tracking of Brain Metastases using Convolutional Neural Networks](#). *61st American Society for Radiation Oncology (ASTRO) Annual Meeting*, Chicago, Illinois (2019).
13. **Houri J**, Connor M, Karunamuni R, McDonald C, Seibert T, White N, Pettersson N, Dale A, Hattangadi-Gluth J, Moiseenko V. [Tract-Crawler: A Computational Tool to Analyze Regional White Matter Dose Effects after Brain RT](#). *37th European Society for Radiotherapy and Oncology (ESTRO) Annual Conference*, Barcelona, Spain (2018). – **Scored in top 13% of abstracts (allocated to oral presentation or poster discussion)**.

INVITED Talks

1. “What You Wear in Space Can Save Your Life: The AstroRad Vest”. *Future In-Space Operations (FISO) Telecon Seminar Series*, Online. June 25, 2025.
2. “Modeling and Shielding Human Space Missions”. *The Pennsylvania-Ohio-West Virginia Chapter of AAPM Fall Symposium*, Pittsburgh, Pennsylvania. September 16, 2023.
3. “Proton CT Reconstruction based on Scatter Rejection”. *Seventh Annual Loma Linda Workshop on Particle Imaging and Radiation Treatment Planning*, Online. August 3, 2021.

INTERVIEWS / PRESS COVERAGE

1. Krywko, Jacek. “[We’ve flown a radiation-blocking vest to the Moon and back, and it worked](#).” *Ars Technica*. August 13, 2026.
2. “[A vest for blocking cosmic rays, and updates on NIH funding](#).” Podcast Interview by Sarah Crespi, host. *Science Magazine Podcast*. August 13, 2026.
3. Dunn, Marcia. “[Radiation vest tested on NASA moonshot can cut astronauts’ exposure in solar storms, study suggests](#).” *Associated Press*. August 12, 2026.
4. Choi, Charles Q. “[Radiation-shielding vest aced Artemis I lunar test, could protect astronauts on moon and Mars missions](#).” *SPACE.com*. August 12, 2026.
5. Ware, Skyler. “[A vest tested on the Artemis I mission blocked some radiation exposure](#).” *ScienceNews*. August 12, 2026.

6. Apte, Poornima. "[ISS-Tested Vest Could Safeguard Astronauts from Solar Particle Events](#)." *The American Society of Mechanical Engineers*. April 8, 2025.
7. Livingston, Stephenie. "[Armor for Astronauts: Developing High-Tech Wearable Radiation Shields](#)." *Upward, the official magazine of the ISS National Lab*. November 13, 2024.